

Manipal Institute of Technology, MAHE



SOFTWARE ENGINEERING LAB
(CSE 3141)

PORTKEY
Food Delivery System

Project Report

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1. Justify whether the RAD model can be applied for your Project. Discuss with key points.

RAD (Rapid Application Development) Model Applicability:

The RAD model emphasizes rapid delivery, extensive user involvement, and iterative prototyping. It works well for projects where user interfaces are crucial and requirements change over time.

Why can PORTKEY use RAD?

- **Module Independence:** Each of the system's clearly defined modules User Authentication, Restaurant Listing, Ordering, Payment, Chatbot, and Delivery can be quickly prototyped separately.
- **UI/UX prototyping:** Iterative prototyping and refinement are best suited for highly visual elements such as restaurant menus, order carts, and tracking user interfaces.
- **Robust User Feedback Loop:** As a food delivery platform, user experience is crucial, and RAD facilitates ongoing stakeholder (customers) participation.
- **Reusability:** Time can be saved by reusing chatbot components, database schema, and APIs throughout iterations.

2. Justify whether Extreme Programming model can be applied for your project. Justify the answer with key points.

Extreme Programming (XP) Applicability:

The principles of Extreme Programming (XP) include frequent releases, pair programming, test-driven development (TDD), and continuous feedback. Small teams and projects with shifting needs are ideal for it.

XP's compatibility with PORTKEY:

- **Small Team Structure:** XP works very well with just three people.
- **Dynamic Requirements:** XP welcomes the anticipated evolution of features like order tracking, chatbot responses, and payment integrations.
- **Test-Driven Development (TDD):** Regular testing is required due to the critical nature of the payment system, order confirmation, and chatbot.
- **The benefit of pair programming** is that two people can work together to produce higher-quality complex modules like chatbot integration/ the payment logic.
- **Continuous Integration:** Lowers deployment risks by promoting regular integration.

3. Discuss whether the SCRUM model can be applied for your project.

SCRUM Model Applicability:

Sprints, roles, and backlog prioritization form the foundation of the Agile framework known as SCRUM. It is perfect for dynamic projects with incremental delivery.

Why PORTKEY can use SCRUM:

Breakdown Based on Sprints:

Sprint 1: Restaurant Listing and Login

Sprint 2: Cart System and Orders

Sprint 3: Integration of Payment Gateways

Sprint 4: Tracking Module & Chatbot

Sprint 5: Testing and Deployment

Flexibility for Changing Requirements: Restaurant modules, payment methods, and chatbot behaviour can all be modified as backlog items.

Every sprint produces a functional version, such as a restaurant listing that is ready before chatbot integration. This is known as incremental delivery.

4. Design the Spiral model for your project.

